L-8

X, Y are independent r.v s

$$\underbrace{P(X = x, Y = y)}_{\text{Joint PMF of } X \,\&\, Y} = P(X = x)\,P(Y = y)$$

for all x, y in the range of X and Y

$$P(X \leq x, Y \leq y) = P(X \leq x)\,P(Y \leq y)$$

Joint CDF of X & Y

$\forall\, x, y$ in $R(X)$ & $R(Y)$

$X_1, X_2, \ldots, X_n$ independent r.v s

$\Leftrightarrow P(X_1 \leq x_1, X_2 \leq x_2, \ldots, X_n \leq x_n)$

$$= P(X_1 \leq x_1) P(X_2 \leq x_2) \ldots P(X_n \leq x_n)$$

$$\forall\ x_1, x_2, \ldots, x_n$$

$$P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)$$
$$= P(X_1 = x_1) \ldots P(X_n = x_n)$$

$$P(X = 2, Y = 3) = P(X = 2) P(Y = 3)$$

3 4

$$P(X = 2, Y = 4) = P(X = 2) P(Y = 4)$$
$$P(X = 3, Y = 3) = P(X = 3) P(Y = 3)$$
$$P(X = 3, Y = 4) = P(X = 3) P(Y = 4)$$

$|R(X)| = 3$
$|R(Y)| = 3$
$|R(Z)| = 3$

i.i.d random variables = independent & identically distributed

X, Y iid r.vs

$X \sim Ber(p)$

$Y \sim Ber(q)$

$X \sim Bin(n, p)$

$X = X_1 + X_2 + \cdots + X_n$

$\downarrow$

$X_i \sim Ber(p) \ \forall \ 1 \leq i \leq n$

X_i are i.i.d r.v.s

① n games are being played, independently

Each game ends in a win for 1 player with $p = 0.4$

& ends in a draw for 1 player with probability 0.6.

(i) Find the distribution of the no. of games ending in a draw

X = # of games ending in a draw

$P(X = k) = {}^nC_k \cdot (0.6)^k (0.4)^{n-k}$

(ii) Find the distribution of the no. of players whose games ended in a draw

Y

$Y = 2X$

$$P(Y=k) = P(2X=k)$$
$$= P(X = k/2)$$

Ex (iii) Show that X & Y are not independent using joint PMF.

2. Suppose X & Y are 2 discrete r.vs s.t
$$P(X=Y)=1$$
(a) Do X & Y have the same PMF?
(b) Is it possible for X & Y to be independent?

(a) PMF of X = PMF of Y
$$P(X=x) = P(Y=x) \quad \forall x$$
$$P(X=x, Y=y) = P(X=x)\,P(Y=y)$$
$\forall\ x, y$ $\quad x \to R(X)$, $y \to R(Y)$

Case 1: $x = y$

Case 2: $x \neq y$

$x \neq y$

$\to 0$

$X \sim \text{Bern}(1/2)$

$Y = 1 - X \sim \text{Bern}(1/2)$

Averages (means, expected value)

$1, 2, 3, 4, 5$ $\qquad \frac{1+2+3+4+5}{5} = \cancel{2.5}\ 3$

$1, 1, 1, 1, 1, 3, 3, 5$ $\qquad \frac{1+1+1+1+1+3+3+5}{8} =$

$$\frac{5}{8}\cdot 1 + \frac{2}{8}\cdot 3 + \frac{1}{8}\cdot 5$$

$$E(X) = \sum_{x:\ P(X=x)>0} x\, P(X=x)$$

$X \sim Bern(p)$

$$E(X) = 1.\ P(X=1) + 0\cdot P(X=0)$$
$$= p + 0\cdot(1-p)$$
$$= p$$

$X \sim Bin(n, p)$

$$E(X) = \sum_{k=0}^{n} k.\ {}^{n}C_{k}\cdot p^{k}(1-p)^{n-k}$$

$$= \sum_{k=1}^{n} n.\ {}^{n-1}C_{k-1}\ p^{k}(1-p)^{n-k}$$

$$= n \sum_{k=1}^{n} {}^{n-1}C_{k-1}\, p^{k-1}\, p\, (1-p)^{(n-1)-(k-1)}$$

$$= np \left(p + (1-p)\right)^{n-1}$$

$$= np$$

$$E(X+Y) = E(X) + E(Y)$$